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INTERIM FEDERAL STANDARD

FLAME-SPREAD PROPERTIES OF MATERIALS

This Interim Federal Standard was developed by the Department of Commerce, National Bureau of Standards, Washington 25, D. C., based upon currently available technical information. This is the third draft of this proposed standard. It embodies all of the technical changes that have been suggested. Also, it contains saving clauses which are intended to prevent misuse or misunderstanding.

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S1. PURPOSE AND SCOPE.

S1.1 Purpose. To provide a method of determining the comparative surface flame-spread properties of various materials, and to establish a standard for flame-spread classification of materials. It has been prepared for use as an alternate for the Tunnel Test, ASTM E84, where such alternate use is desired by the Federal procuring agency. Such use of the standard should be made with caution since it should not be assumed that identical flammability classifications will result, even though there appears to be a general correlation between the classifications by the two methods. Also, for other than Federal purchasing, users of this standard should obtain assurances that such use is not in contravention of local ordinances.

S1.2 Scope. The method is applicable to materials that may be used in such ways as to provide exposed surfaces over which the flames of accidental fires may spread. These include materials in the form of boards, sheets, structural members and certain heavy fabrics; finish materials applied as liquids, films or sheets, or combinations thereof, with or without intermediate adhesive substances.

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S2. DESCRIPTION OF TEST METHOD[1].

S2.1 The method employs a radiant heat source consisting of a 12- by 18-inch panel in front of which an inclined 6- by 18-inch specimen of the material under test is placed. The orientation of the specimen is such that ignition is forced near the upper edge of the specimen and the flame front progresses downward. Figure 1 illustrates the assembled test equipment in use.

S2.2 A factor for the progress of the flame front and a factor which is related to the rate of heat generation by the material under test are combined by means of an empirical relationship to provide a flame-spread index for the material.

S2.3 The method includes means for the quantitative determination of the amount of smoke evolved during the test. While not a factor in the determination of the flame-spread index, this indicates the extent of a parallel hazard.

S2.4 The results obtained by use of the test method in the prescribed manner are relatively unaffected by slight changes in ambient variables and modifications in mechanical adjustment. The classification of materials with respect to surface flammability, derived by the method, appears to be generally consistent with information available on the behavior of such materials during fires. For some materials, there is evidence of a limited correlation between the results of this test method and the tunnel test method ASTM E84-61. However, flame-spread index values by this test method should not be assumed to be numerically equivalent to flame spread classifications by the tunnel test method nor should one test method be considered as an identical or automatic alternate for the other. In the preparation or revision of an end-product Federal Specification, the applicable test method, if any, must be specified.

S3. CLASSIFICATION OF MATERIALS BY FLAME-SPREAD INDICES

S3.1 The classification of materials shall be based upon the mean value from a minimum of four specimens according to the ranges given below for flame-spread indices determined by this method. The flame-spread index is computed such that a higher index corresponds to a higher degree of flammability, all measurements being made relative to 0 for incombustible asbestos-cement board.

[1]Factors in the development of the test method are described in "A Method for Measuring Surface Flammability of Materials Using a Radiant Energy Source" by A. F. Robertson, D. Gross, and J. Loftus, ASTM Proceedings, Vol. 56 (1956).

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S3.1.1

Classification*	Range of Indices
I	0 - 15
II	16 - 25
III	26 - 75
IV	76 - 200
V	Over 200

*Modifier with regard to flash, running, or dripping to be included, when applicable. See Sections S6.2 and S6.3.

S3.2 The following table gives the mean flame-spread indices of some representative materials as determined by the method of this standard. The data are provided to permit comparison of test results with those of familiar materials.

S3.2.1

Material	Mean Flame-Spread Index I_s	Smoke Deposit mg	Class
asbestos-cement board	0.0	0.0	I
aluminum foil over plywood*	1.0	0.1	I
gypsum board	5- 20**	0.1	I, II
paint, fire-retardant on plywood*	20- 50**	1.0-1.2	II, III
paint,***on plywood*	50-100	0.4-1.7	III, IV
fiberboard, factory finished	80-110**	0.2-0.5	IV
red oak	80-160**	0.3	IV
hardboard	120-160**	2.0-5.0	IV
plywood*	120-200**	0.3-0.8	IV
fiberboard, unfinished	>200	0.2	V
shellac on plywood*	>200	0.2	V

* Exterior grade Douglas fir.

** Indicates variations observed among mean values of similar materials but of different density, orientation, lot or source.

*** Oleoresinous, alkyd and latex.

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S4. APPARATUS. The apparatus shall be installed in a room, laboratory or test area in such manner as to be free from strong drafts which could disturb the pattern of burning on the specimen surface or the measurement of its heat evolution. No restrictive side walls or tables, highly reflective surfaces or windows shall be placed in front of the plane of the radiating surface and within 3 feet of the radiant panel or specimen.

S4.1 The test equipment shall consist of a radiant panel with its associated gas and air supply, a specimen holder and framework for its support, a stack fitted with thermocouples, a smoke sampling assembly (the positioning of these parts is shown in Figure 2); also a pilot burner, a radiation pyrometer, a potentiometer indicator or recorder, a recording potentiometer and a timer. In addition, a hood and a conditioning room should be provided.

S4.1.1 Radiant panel[2]. A porous refractory material enclosed, on all but one face, in a cast iron frame, vertically mounted. The overall dimensions are 19- by 13- by 4- inches, the vertical refractory face 18- by 12- inches. The panel is equipped with a venturi-type aspirator for mixing gas supplied at approximately atmospheric pressure with air. The air is supplied by a centrifugal blower with rated capacity of 100 cfm at 2.8 inches water pressure, or equivalent. An air filter is provided to prevent dust from obstructing the panel pores. A pressure regulator and a control and shut-off valve are provided for the gas supply.

S4.1.2 Specimen holder and framework. Shown in Figures 2 and 3. Holder preferably fabricated of a heat-resistant steel. The specimen holder is placed on a metal framework which is arranged for maintaining the correct spacing and for positioning of the specimen centrally with respect to the radiant panel. The framework dimensions must be maintained to within 1/8 inch of the values given in Figure 2.

S4.1.3 Pilot burner. Used to force ignition, when possible, at the top of the specimen. The burner consists of a short length of 3/16-inch O.D., 1/8-inch I.D. stainless steel tubing the exposed portion of which is protected by a 9/32-inch O.D. porcelain tube. The pilot burner is fed with acetylene premixed with air. The air may be supplied from any suitable mixing

[2]Type 1 surface combustor, 12- by 18-inches, and associated air and gas supply system, manufactured by Radiant Heat Ltd., London N. 1, England, and distributed through their export agent Standard Sales,Ltd., 74 Borough High St.,London S.E.1, England.

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device such as an aspirating-type fitting. To reduce the frequency of replacement of the pilot burner from oxidation effects, it is mounted to swing out of position readily when not in use. When in use, the burner is horizontal and at a slight angle with respect to the specimen. The position of the burner tip is such that a minimum of 2 inches of the burner flame is in contact with or not more than 1/2 inch from the central uppermost area of the exposed surface of the specimen.

S4.1.4 Stack and thermocouples. -Dimensions of the vertically-mounted stack, locations for the eight thermocouples therein, and position of the stack with respect to the specimen and radiant panel are shown in Figure 2. The thermocouples, of 0.020-inch chromel and alumel wires, are supported within porcelain insulators as shown in Figure 4. The thermocouples are matched for equal resistance and connected in parallel to give a single output indicative of the average of the temperatures at the eight junctions. This output is recorded by an automatic recording potentiometer (see S4.1.6).

S4.1.5 Smoke sampling device. -A means for drawing air and gases from the top of the stack through a glass fiber filter paper.[3] The details are shown in Figure 5. A flow meter or other gas metering device and an aspirator or vacuum pump are used to maintain a constant velocity of flow over the 7/8-inch diameter exposed filter disc equivalent to 40 feet per minute (20.4 cm per second) of air at 70°F.

S4.1.6 Recording potentiometer. -Range: 100 to 1000°F. The recorder should give a continuous record or should print every 15 seconds or less.

S4.1.7 Radiation pyrometer. -Used for standardization of the radiant output of the panel. The device should be suitable for viewing a circular area of 10-inches diameter at a convenient range. It is preferable that a wide angle radiation pyrometer with high transmission in the infrared, such as one employing a calcium fluoride lens or window, be used.

[3]Type 1106 BH All-Glass Filter Media, distributed by Mine Safety Appliances Co., 201 North Braddock Ave., Pittsburgh 8, Pennsylvania, or equivalent.

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S4.1.8 Potentiometer indicator or recorder. Manual or automatic, to monitor the electrical output of the radiation pyrometer. It should have a range suitable for indicating the output of the radiation pyrometer at the operating blackbody temperature (see calibration procedure S5.1.3) and when used with the radiation pyrometer should have a sensitivity and limit of error corresponding to an error in the blackbody temperature of ± 2°C.

S4.1.9 Timer. -Should have start, stop and reset features and should be suitable for rapid visual reading. The timer should be calibrated up to at least 15 minutes, preferably in minutes and hundredths.

S4.1.10 Conditioning room. A room or chamber of sufficient size to hold several specimens and controlled to provide an ambient of 73 ± 5°F and 50 ± 5 percent relative humidity. It should be sufficiently close to the test apparatus that the transport of each individual test specimen between the two may be made readily. Vapor barrier jackets in which specimens may be placed for this transport should be available.

S4.1.11 Hood. For standardization of the smoke measurement, a hood with exhaust blower placed above the stack is required. The blower should be capable of producing a velocity of 100 feet per minute at the top of the stack with the radiant panel not operating. This corresponds to approximately 250 feet per minute with the radiant panel at operating temperature. The velocity through the stack is not critical for flame-spread measurements provided a stack thermocouple temperature calibration is performed for the established test conditions (see 5.1.4). The hood surfaces should clear the top and sides of the stack by a minimum of 10 inches and 7 1/2 inches respectively.

S4.1.12 Smoke Transmission Densitometer. A photometer using an S-4 type photosensitive surface together with an incandescent light source shall be used for optical density measurements of the deposited smoke film over a density range of 0 to 4.5.

S5. PROCEDURES. The operational procedures to be followed with the apparatus described are divided into two general groups: 5.1) those necessary for the calibration and standardization of the apparatus, and 5.2) those necessary for the performance of a test of a specimen.

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S5.1 Calibration procedures. In setting up the apparatus, initially, several precautions should be observed and calibrations made.

S5.1.1 Alignment. Considerable care should be taken that the various components are well supported, aligned, and at the correct distances from one another, as given in Figure 2. Stops, guides or rests should be provided so that the movable parts and those that require periodic cleaning or replacement, may be moved into normal operating positions easily and accurately. These include the pilot burner, smoke sampler, radiation pyrometer, stack, and specimen holder.

S5.1.2 Smoke sampling. The smoke sampling device, including all connections should be tested by substituting an impermeable diaphragm for the filter disk. With the vacuum pump or aspirator in operation, the flowmeter should indicate zero air flow. Since the gas flow resistance of the filter disk is not negligible, leaks would result in erroneously low smoke deposits on the disk and should be eliminated.

Allowance for the equilibrium moisture content of the glass fiber filter disks should be made in the smoke deposit measurements. During the test of an asbestos-cement board specimen, the loss in weight of the filter disk should be recorded. This weight loss should be applied as a correction to the smoke deposit measurements obtained in tests of materials. (see S5.2.10.)

S5.1.3 Radiation pyrometer. The radiation pyrometer should be calibrated at the operating point, i.e., a blackbody temperature of 1238°F (670°C), and its approximate temperature-emf relation in this vicinity determined. This provides data on a segment of the calibration curve necessary for operation within the limits set. (see S5.2.2.) The calibration consists in measuring the emf produced when the pyrometer is sighted on a blackbody source, the temperature of which is measured by means of a calibrated thermocouple. A suitable blackbody enclosure may consist of a hollow, rough-surfaced chromel cylinder closed at one end and with a sight hole in the other end, placed within a furnace and maintained at a uniform temperature. A 20:1 ratio of the total area of the internal surface of the blackbody to the area of the blackbody opening is considered adequate. The radiation pyrometer is placed so that its field of view is confined to the opposite end of the cylinder where a thermocouple inserted within a drilled hole in the body wall and extending almost to the base of the cylinder indicates the blackbody temperature.

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The proper position for the radiation pyrometer when monitoring the output of the radiant panel is along the normal through the center of the panel surface, and at the distance to view a central circular area of 10 inches diameter on the panel. This position should be determined as follows: With the panel operating and the pyrometer sighting along the normal, slowly insert an incombustible opaque shutter in front of the panel from top, bottom and each side in turn, noting how far the shutter must be inserted before the pyrometer output reading starts to decrease. Adjust the pyrometer along the normal until the position is found at which the output decrease starts when the shutter is 1-inch from each side and 4-inches from the top or bottom. The pyrometer should be fixed in this position on a sturdy mount.

S5.1.4 Stack Thermocouples. With the panel at operating temperature, and the exhaust blower producing an established stack velocity, note the temperature of the stack thermocouples. The operating stack thermocouple temperature is dependent upon the position of the hood exhaust system opening with respect to the stack. Although flame-spread measurements may be readily made for a wide range of operating stack thermocouple temperatures (i.e. prior to placing test specimen in position), difficulties with smoke measurements have been noted in cases where this temperature was significantly greater than 450°F (232°C). For this reason it is recommended that initial positioning of the hood exhaust system be made so as to maintain the operating stack thermocouple temperature within the range 350-450°F (177-232°C).

Place an asbestos-cement board specimen, with a 1/2-inch asbestos millboard backing, in test position, and note the ensuing equilibrium temperature of the stack thermocouples which will be used as a base temperature for the following procedure.

A multi-ported diffusion (no premixed air) burner shall be prepared from a 12- to 15-inch length of 1/4-inch standard wrought iron or steel pipe capped at one end and containing ten 0.070 inch diameter radial holes spaced 5/8 inch on centers along a line parallel to the axis of the pipe. Place the centerline of the pipe burner in a horizontal position 1 inch (measured along the specimen surface) below the upper exposed edge of the asbestos-cement board specimen. The pipe wall shall be in contact with both side edges of the specimen holder so that the portion of the pipe containing the burner holes is centered with respect to the specimen. The axes of the burner holes shall be

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vertical causing flames from the burner to impinge at or near the top of the asbestos-cement board specimen. The type and orientation of the yellow diffusion flames produced are comparable to the flames emitted from a burning specimen. Record the maximum stack thermocouple temperature rise above the previously defined base for each of several gas flow rates to the burner allowing a minimum of 10 minutes at each flow rate for stack temperature stabilization. The gas supplied to the calibration burner shall be manufactured, methane, or natural gas or combinations of these gases. The gas flow rate to the calibration burner should be measured by means of a calibrated flow meter. Use the higher (gross) heating value of the gas to convert the gas flow rates to heat input rates. Moisture, temperature and pressure corrections should be applied, when applicable, to convert the gas flow rates and the higher (gross) heating value of the gas to a dry basis at a standard temperature of 60°F. and a standard pressure of 30.0 inches of mercury. Plot the maximum stack thermocouple temperature rise, in degrees Fahrenheit, as a function of the corresponding measured heat input rate in Btu per minute. The slope of the line fitted to these points is the value of β in the flame-spread index formula. (see S6.1.) If changes are made which affect the velocity through the stack, a new value of β must be determined for these new conditions.

S5.2 Test procedure.

S5.2.1 All specimens, prepared in the 6- by 18-inch size, should be pre-dried for 24 hours at 140° F and then conditioned to equilibrium with an ambient of 73 ± 5° F and 50 ± 5 percent relative humidity.

Since the base material (to which the finish material is applied) is an important factor in the flame-spread behavior of composite structures with thin finish materials (ref. ASTM Bulletin No. 230, pp. 56-60, May 1958), the following procedures of specimen preparation should be observed:

S5.2.1.1 Where the intended applications of finish materials to base materials are known, the specimen shall be of the intended assembly, and be prepared by the intended application procedures.

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S5.2.1.2 Where the intended application of a finish material is not specified, or may be any of several, the finish material may be prepared for test as follows:

a) Opaque (to infrared radiation) sheet materials of greater than 1/16-inch thickness are not applied to any base material.

b) Opaque sheet materials of 1/16-inch thickness or less shall be applied to 1/4-inch thick tempered hardboard, using recommended application procedures. The hardboard base material alone shall have a mean flame-spread index within the range 130 to 160 based upon a minimum of four tests performed according to this test method.

c) Liquid films, such as paints, and other materials intended for application to combustible base materials, shall be applied to the smooth surface of 1/4 inch thick tempered hardboard, as defined in b), using recommended application procedures.

d) Liquid films, such as sealers, adhesives, etc., and other materials intended for application to non-combustible base materials or being tested for their inherent flame-spread contribution shall be applied to the smooth surface of 1/4-inch thick asbestos cement board. In the absence of specific thickness or spreading rate requirements, the coating shall be built up to a thickness of at least 0.030 inch.

e) Diathermous (transparent or translucent) sheet materials of any thickness are not applied to any base material but are backed by a sheet of highly reflective aluminum foil.

S5.2.1.3 Materials, including fabrics, that will not be applied to any base material, but will be supported only at one or more edges, shall be mounted on a special backing. This backing shall consist of a sheet of 1/2-inch thick asbestos millboard one face of which is covered by a sheet of highly reflective aluminum foil with 1/2- by 1/2-inch spacers of asbestos millboard along top, bottom, and both edges of the foil-covered face. The test material shall be placed across the

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spacers, thereby leaving a 1/2-inch space between the test material and the aluminum foil surface. Flexible materials shall be cut in sheets approximately 10 by 22 inches, the edges folded around the frame, and fastened to the rear surface of the millboard. Such materials shall be pulled tight enough to eliminate slack, but no tighter.

S5.2.1.4 Materials, including fabrics, which tend to crack and fall, peel, run or drip or otherwise separate from the specimen holder during the test exposure shall be retained in position by means of a 6- by 18-inch sheet of 1-inch hexagonal mesh poultry netting placed in the specimen holder prior to the insertion of the test material.

S5.2.1.5 Finish materials including sheet laminates and tiles which are applied to a base material with glue or adhesive as well as sheet laminate materials not glued to any base should be tested for the possibility of increased flame-spread or associated hazard due to delamination, buckling, separation, peeling or falling of the finish material. Increased flame-spread may be due to flaming on the reverse side of the finish material, on the adhesive, or on the base material. To test for such behavior, prepare one specimen with grooves 1/16 inch or less in width scored in the finish material as follows: one longitudinal groove 1 inch from the edge and five lateral grooves 4 inches apart starting 1 inch from the top (or bottom) edge. For laminates not glued to a base material, the grooves should be scored through three-fourths of the material thickness, while for glued materials, the grooves should be scored completely through the finish material. This yields a pattern of four 4-inch tiles supported by continuity on one edge and by the internal layers of the laminate or by adhesive.

Compare the results of a flame-spread test on this specimen with those on a conventional specimen, i.e., one without cut grooves.

For those assemblies in which the adhesive is not adversely affected by the test exposure, or in which no appreciable delamination results, the magnitude of the differences between the test results on the conventional and grooved specimens will be on the order of the experimental error. In such cases, or if an increased flame-spread index is attributable primarily to increased flame travel within the grooves, the remaining specimens shall be tested in the conventional manner.

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For those assemblies in which the adhesive is affected by the test exposure, or in which appreciable delamination occurs, determine whether the grooved specimen does increase the flame spread or associated hazard. If it does, comparison should be made between the results of the grooved specimen and an ungrooved specimen retained in position in the specimen holder by 1-inch hexagonal mesh poultry netting as in 5.2.1.4. The specimens remaining after these comparisons have been made shall be tested under the arrangement yielding the highest flame-spread index.

S5.2.2 The gas-air mixture passing through the radiant panel is ignited and the panel allowed to heat for 1/2-hour, minimum. At the end of the preheat period, the radiant output is checked with the radiation pyrometer, positioned to view the central panel area of 10-inches diameter. The gas supply is adjusted to maintain the radiant output at that which would be obtained from a blackbody of the same dimensions and at a temperature of 1238 ± 7°F (670 ± 4°C). (See 5.1.3.)

S5.2.3 Turn on and standardize the recording potentiometer for measurement of the stack thermocouple temperature rise.

S5.2.4 Weigh a new filter disk to an accuracy of .0001 gm. and place it in the smoke sampling device.

S5.2.5 After steady-state conditions at the stack have been reached, remove one specimen from the conditioning room and bring to the test apparatus. Either of two procedures may be followed:

a) Mount specimen in a cool specimen holder in the conditioning room and bring the assembly to the test apparatus in a vapor barrier jacket, or

b) Wrap the specimen in a vapor barrier jacket, bring to the test apparatus, remove from the jacket and quickly mount in cool specimen holder.

In either case, all specimens except those over 3/4-inch thick should be backed with 1/2-inch thick asbestos millboard of 60 lb/ft^3 density. In order to prevent flaming on the back surface of the specimen, a 1- by 6-inch strip of heavy asbestos

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paper (about 1/16-inch thick) is placed across the top edge of the sample and the asbestos millboard backing, and is folded down over the back face of the millboard, to form a seal between the millboard and specimen holder.

NOTE: The specimen should be tested as quickly as practicable after having been removed from the conditioning room, in order to minimize any changes in the moisture content and temperature of the specimen attained in the conditioning room.

S5.2.6 Place the smoke sampling device in position above the stack and adjust the gas flow through it (see S4.1.5); maintain this gas flow, if possible, by adjusting the suction throughout the test.

S5.2.7 Ignite the pilot burner, and adjust the flame to 2- to 3-inches length. Swing the pilot burner into operating position and maintain for the duration of the test. If, during the test, the pilot burner flame is extinguished, relight. For materials which tend to shrink or contract upon application of heat, the pilot burner tip should be turned toward the specimen so that more direct flame impingement on the specimen is assured.

S5.2.8 Place the specimen holder containing the specimen in test position and start the timer simultaneously. A single operator may accomplish this conveniently by means of a foot-operated switch on the timer as shown in Figure 1. Make continuous observations of the development and spread of flames on the specimen surface, recording both the elapsed times at which any portion of a sustained flame front arrives at points marked 3, 6, 9, 12, and 15 inches from the upper edge of the specimen and important visual observations.

S5.2.9 Continue the test until the flame front shall have progressed the full length of the exposed specimen or until 15 minutes, whichever is lesser. Discard the specimen and allow the specimen holder to cool.

S5.2.10 Carefully remove and reweigh the smoke sampling filter disk. Making allowance for the loss in weight of the filter disk at the operating temperature (see S5.1.2), determine the weight of the smoke deposit to the nearest 0.0001 gm.

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S5.2.11 After weighing, the smoke sampling filter disk shall be measured for optical density with a transmission densitometer (see S4.1.12). Likewise, the clear peripheral area of the filter disk shall be measured for optical density.

S5.2.12 Determine the maximum stack thermocouple temperature from the automatic recording potentiometer.

S5.2.13 Determine the mean temperature measured by the stack thermocouples over the final 10 minutes of a 15 minute interval using an asbestos-cement board specimen with pilot burner in position. The difference between this temperature and the maximum stack thermocouple temperature (see S5.2.12) is the value of $\Delta\theta$ for the test specimen.

S5.2.14 For low density, cellular or other materials in which the flaming is rapid and is limited to the very early part of the test exposure, a slight temperature rise may go undetected due to the response time of the stack thermocouples. Provided the stack thermocouple temperature does not exceed that measured in S5.2.13 by more than 10 degrees F, the value of $\Delta\theta$ for the test specimen shall be determined with respect to the stack thermocouple temperature for the asbestos-cement board specimen at a time corresponding to that at which the maximum stack thermocouple temperature was observed for the test specimen . Continuous recording of the stack thermocouple temperature is required in such cases.

S5.3 Maintenance. In addition to the normal operating procedures, various procedures should be carried out as required.

S5.3.1 Radiant panel. The air intake filter should be checked occasionally and cleaned or replaced as needed to prevent the passage of dust which might obstruct pores in the refractory.

S5.3.2 Stack thermocouples. Should be inspected and brushed off periodically since heavy deposits of combustion products reduce the response and accuracy of the temperature measurements. The thermocouples should be cleaned no less frequently than after 40 hours of use. With very heavy smoke-producing materials, it may be necessary to clean the thermocouples after each test.

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S5.3.3 Smoke sampling device. Should be checked for leaks periodically. (See S5.1.2)

S5.3.4 Position checks. The positioning, by stops, guides, etc. of movable components (specimen holders, smoke sampling device, radiation pyrometer, etc.) should be checked periodically.

S5.3.5 Calibration checks. The calibrations of all measuring instruments, meters, potentiometer and other auxiliary equipment should be checked periodically.

S5.3.6 Pilot burner. Should be replaced or cleaned periodically to remove deposits formed on the interior surface.

S6. CALCULATIONS.

S6.1 Flame-spread index. The flame-spread index, I_s, is taken as the product of the flame-spread factor, F_s, and the heat evolution factor, Q, thus

$$I_s = F_s\, Q$$

where $F_s = 1 + 1/t_3 + 1/(t_6-t_3) + 1/(t_9-t_6) + 1/(t_{12}-t_9) + 1/(t_{15}-t_{12})$.

In which t_3, t_6, t_9, t_{12}, and t_{15} are the elapsed times in minutes, from the start of specimen exposure until arrival of the flame front at distances from the top of the specimen indicated by the numerical subscripts. Certain materials, or specimens, may show local advance more rapid than the general flame front advance not due to melting and running or dripping of burning material, but caused by non-homogeneity of the material (local fissure, combustible-rich segment, etc.). The times associated with the furthest advance, whether local or general, shall be used in computing the flame-spread factor F_s.

For some materials, particularly those with flame-retardant treated surfaces, a significant delay may occur in the start of surface flaming, followed by a very rapid flame progression which may envelop one or more markings before a sustained flame establishes itself. In many cases, the progression of the sustained flame front after this has occurred, will be regular and

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times for the flame front to pass succeeding 3-inch marks may be readily measured. However, in some cases, the sustained flame front progression will be limited, providing no additional time-distance data. If the rapid flaming envelops two or more markings, the time interval(s) involved, even if measurable, would result in a disproportionately high number being used in summing up the components of the flame-spread factor. In order to obtain more representative factors in such situations, the following procedures shall be used:

(1) Measure or estimate the distance D, in inches, and the corresponding time t, in minutes, when the flaming establishes very rapidly past the 3-inch mark or past the 3- and 6-inch marks or past the 3-, 6-, and 9-inch marks.

(2a) In the case where succeeding time-distance data are available, plot D on rectangular coordinates as a function of t on logarithmic coordinates.

(2b) In the case where no succeeding time-distance data are available, determine the distance D_o corresponding to a time of 1.0 minutes from the following equation, $D = D_o + 6.5 \log_e t$.

(3a) Extrapolate the curve (or line) back to obtain estimated times for the 3-in. or 3- and 6-in. markings.

(3b) Using the value of D_o and the same equation from (2b), calculate the corresponding times for distances of 3-, 3- and 6-, or 3-, 6-, and 9-inches.

(4) Use these extrapolated or calculated times and corresponding distances to calculate the flame-spread index.

The heat evolution factor Q, is calculated according to the relation:

$$Q = 0.1\Delta\theta/\beta$$

in which 0.1 is a constant, $\Delta\theta$ is the observed maximum stack thermocouple temperature rise at any stage of combustion of the specimen, in degrees F, over that observed with an asbestos-cement board specimen, and β is the maximum stack thermocouple temperature rise for unit heat input rate to the calibration burner, degrees F min per Btu.

S6.2 Surface flash. The relationship defining the flame-spread factor, F_s, presumes sustained flaming along a portion

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or the entire width of the specimen. In those tests in which flames flash across all, or a significant portion of the surface without immediately establishing sustained flaming over that portion, this relationship would be difficult to apply. In such cases, therefore, if the flame front progresses over at least 6 inches of the length of the specimen in less than a second, the material shall be reported as "Flash Potential" and the time at which the flash occurred shall be reported. If sustained flaming occurs or continues after flashing, I_s is computed in the normal manner, and should be reported in addition to "Flash Potential."

S6.3 Running or dripping of flaming material. -Certain materials may have a tendency to exhibit a fairly rapid running and/or dripping of flaming material either separately or in combination with a general flame front advance. When such running behavior is due to thermal softening or melting aggravated by the near vertical inclination of the specimen, this fact should be noted as "Running of Flaming Material" or Dripping of Flaming Material", and the time at which this running occurred shall be reported in addition to the flame-spread index computed on the basis of the sustained general flame front advance.

S6.4 Smoke deposit. -From the measured weights of the filter disk before and after the test, making correction for any loss in weight due to moisture in the disk, determine the weight of the smoke deposit to the nearest 0.0001 gm. After weighing, the optical density of the deposited smoke film shall be determined by subtracting the average optical density of the clear filter paper disc around its periphery from the average optical density of the filter disc with deposited smoke.

S7. REPORT. The report by the testing laboratory shall include the following:

A. Identification of the material including data such as density, thickness, and for liquid films the number of coats and spreading rate in ft^2/gal.
B. Type of base material used, if any.
C. Number of specimens tested.
D. Flame-spread indices of the individual specimens, and the mean value.
E. Classification according to table in S3.1.1 including applicable modifiers, if any.
F. Mean smoke deposit weight and optical density when measurements of the latter are within the range given in S4.1.12.
G. Important visual test observations of the individual specimens.

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S8. NOTES

S8.1 The flame-spread index, as measured by this test, applies only to the specimen surface which is exposed. In the case of a material the opposite surfaces of which are of different finish and the material is to be used for applications in which either or both surfaces may be exposed, it may be necessary to obtain the flame-spread index for each surface.

S8.2 The applicability of this test method to materials having surface corrugations greater than 1/2 inch depth has not been determined.

S8.3 For checking operational procedural details of this standard, a surface flammability standard (No. 1002) is available, at nominal cost, through the Standard Sample Office, National Bureau of Standards, Washington 25, D. C. The use of this standard material does not obviate the need for following the calibration and standardization procedures outlined herein.

S8.4 The proper calibration of the radiation pyrometer at a blackbody temperature of 1238°F(670°C) as described in 5.1.3 is extremely important. Where facilities for performing such a calibration are not available to laboratories equipped with the radiant panel test apparatus, a check calibration may be performed upon request to the Fire Research Section, National Bureau of Standards, Washington 25, D. C.

S8.5 This standard offers to researchers, testing laboratories, and suppliers of materials for which flame-spread properties are needed, the design and operating procedures of a small, simple, and relatively inexpensive apparatus with which they might become equipped and in a position to make comparative tests as necessary, and for other purposes.

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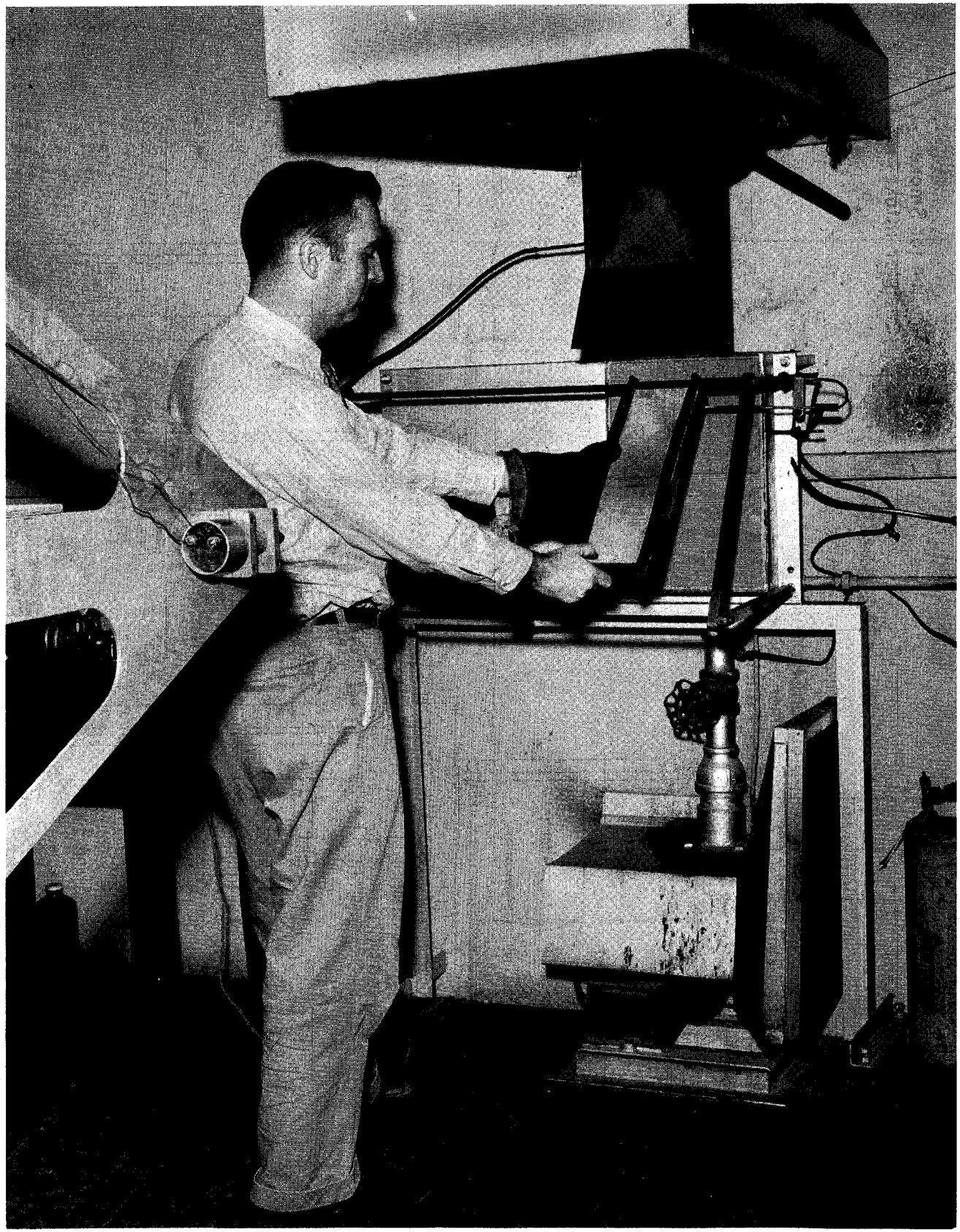


Figure 1. Typical Test Equipment Installation

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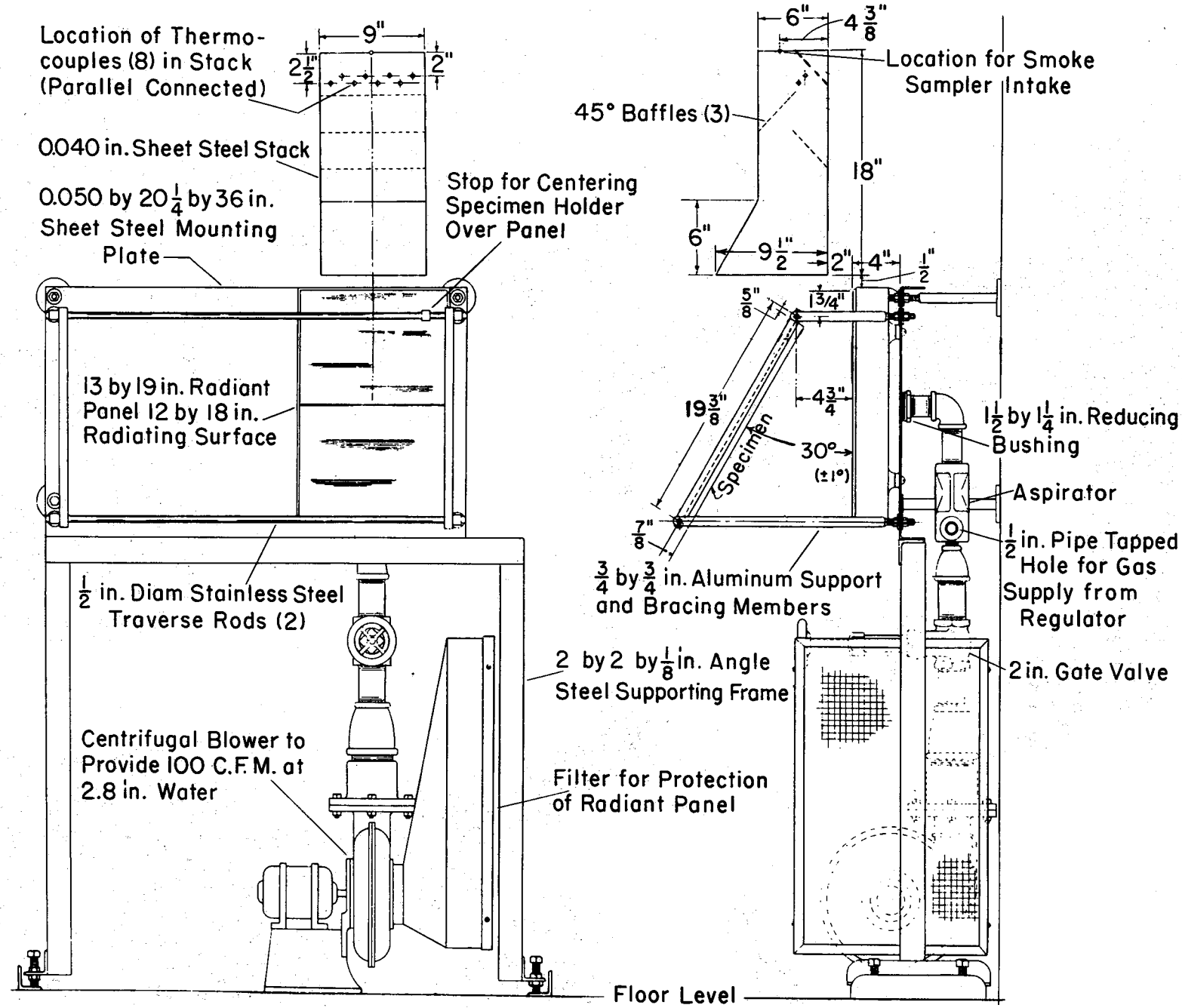


Figure 2 - General Arrangement Drawing

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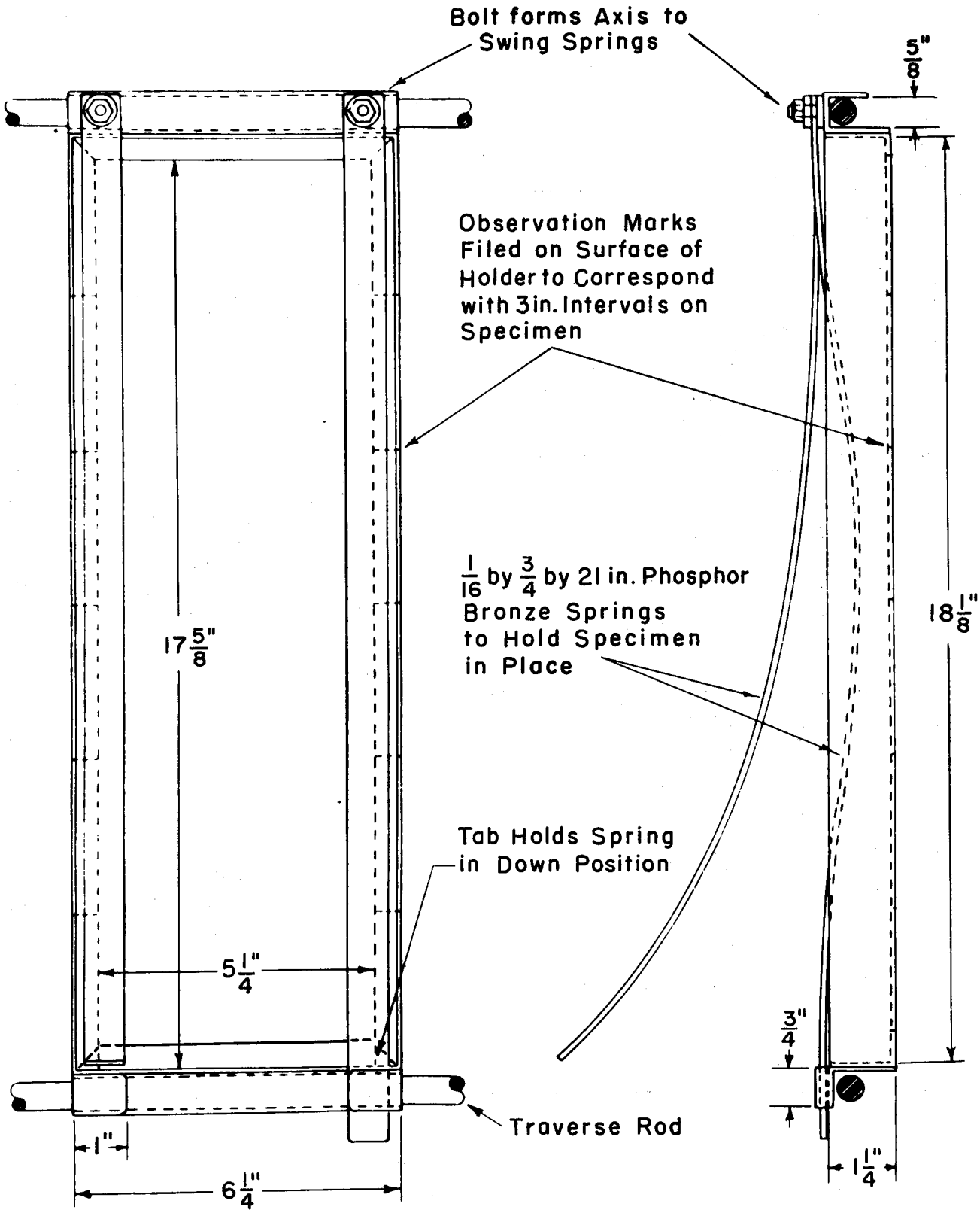


Figure 3. Specimen Holder

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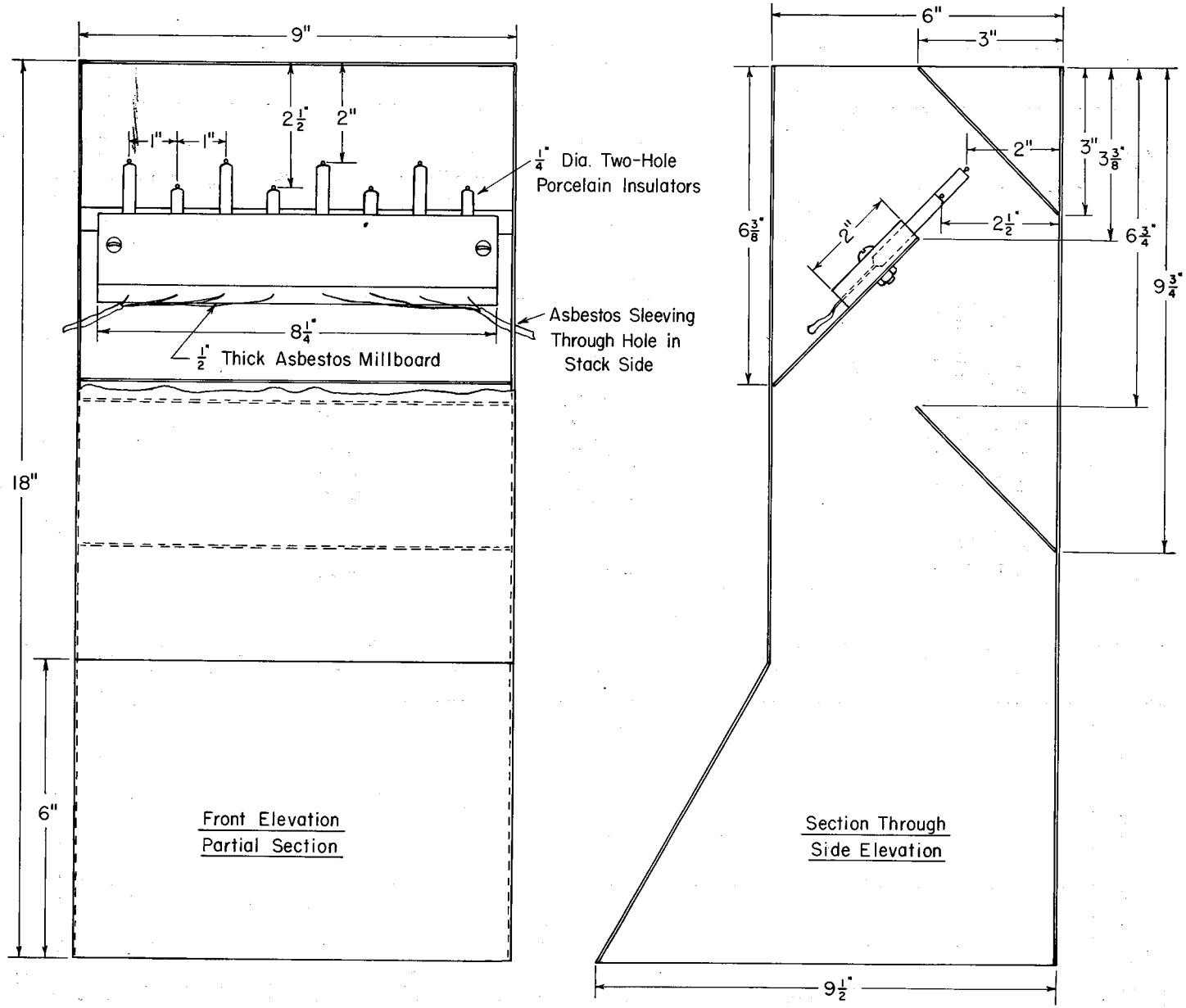


FIG. 4 – STACK THERMOCOUPLE AND BAFFLE ARRANGEMENT

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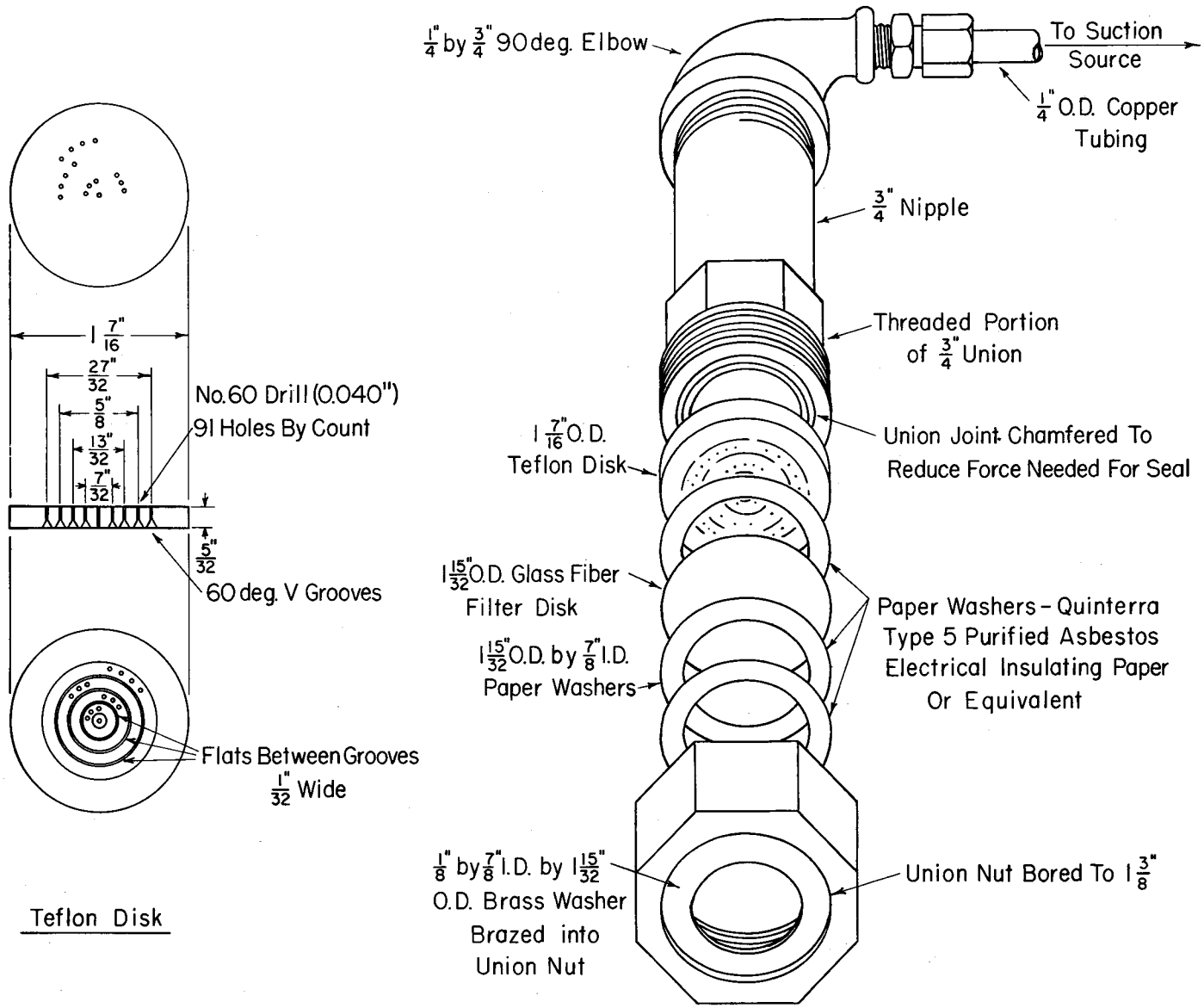


FIG. 5 - SMOKE SAMPLING DEVICE

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